Data Cleaning on Audible Data

Charles Alders

[Audible Dataset on Kaggle (https://www.kaggle.com/datasets/snehangsude/audible-dataset?select=audible_uncleaned.csv)](https://www.kaggle.com/datasets/snehangsude/audible-dataset?select=audible_uncleaned.csv)

This data was scraped from Audible and ended up pretty dirty. Here are the important parts that I cleaned, as seen below:

- The author column has no spaces, and has "Writtenby:" in front of each author's name.
- The narrator column has no spaces, and has "Narratedby:" in front of each narrator's name.
- The time column is text, so I will need to convert it to numeric only. Also, time is somewhat misleading, so I called it 'duration' and counted in minutes instead of hours and minutes.
- The releasedate column is relatively clean, but it was not initialized as a datetime.
- The languages have some inconsistencies in capitalization.
- Stars contains two columns worth of data - rating and amount of ratings. This column also contains unnecessary text, "out of 5 stars".
- Finally, the price column is in Indian Rupees, so I will convert it to USD.
- Other cleaning was done as necessary, such as deleting duplicate rows.

In [1]:
```
import pandas as pd

df = pd.read_csv("audible_uncleaned.csv", index_col=0)
original_df = pd.read_csv("audible_uncleaned.csv", index_col=0)
df.head()
```

Out[1]:

name	author	narrator	time	releasedate	language	sta
Geronimo Stilton #11 & #12	Writtenby:GeronimoStilton	Narratedby:BillLobely	2 hrs and 20 mins	04-08-08	English	5 out stars ratin
The Burning Maze	Writtenby:RickRiordan	Narratedby:RobbieDaymond	13 hrs and 8 mins	01-05-18	English	4.5 o stars ratin
The Deep End	Writtenby:JeffKinney	Narratedby:DanRussell	2 hrs and 3 mins	06-11-20	English	4.5 o stars ratin
Daughter of the Deep	Writtenby:RickRiordan	Narratedby:SoneelaNankani	11 hrs and 16 mins	05-10-21	English	4.5 o stars ratin
The Lightning Thief: Percy Jackson, Book 1	Writtenby:RickRiordan	Narratedby:JesseBernstein	10 hrs	13-01-10	English	4.5 o stars1 ratin

In [2]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 87489 entries, Geronimo Stilton #11 & #12 to Havana
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   author       87489 non-null  object
 1   narrator     87489 non-null  object
 2   time         87489 non-null  object
 3   releasedate  87489 non-null  object
 4   language     87489 non-null  object
 5   stars        87489 non-null  object
 6   price        87489 non-null  object
dtypes: object(7)
memory usage: 5.3+ MB
```

All columns are objects, but I need to change the:

- time to integer

- releasedate to datetime
- stars to integer
- new ratings column as integer
- price to float

In [3]:
```
df.isna().sum()
```

Out[3]:
```
author         0
narrator       0
time           0
releasedate    0
language       0
stars          0
price          0
dtype: int64
```

No missing values! There are still some "missing" values, such as "Not rated yet", but I will handle those later in the cleaning.

In [4]:
```
df.rename(columns={"releasedate": "release_date"}, inplace=True)
```

In [5]:
```
df["author"].replace("Writtenby:", "", regex=True, inplace=True)
df["author"] = df["author"].str.replace( r"([A-Z])", r" \1", regex=True).str.s
```

In [6]:
```
df["narrator"].replace("Narratedby:", "", regex=True, inplace=True)
df["narrator"] = df["narrator"].str.replace( r"([A-Z])", r" \1", regex=True).s
```

Separated the author and narrator values at the capital letter.

In [7]:
```
df.head(10)
```

Out[7]:

name	author	narrator	time	release_date	language	stars	price
Geronimo Stilton #11 & #12	Geronimo Stilton	Bill Lobely	2 hrs and 20 mins	04-08-08	English	5 out of 5 stars34 ratings	468.00
The Burning Maze	Rick Riordan	Robbie Daymond	13 hrs and 8 mins	01-05-18	English	4.5 out of 5 stars41 ratings	820.00
The Deep End	Jeff Kinney	Dan Russell	2 hrs and 3 mins	06-11-20	English	4.5 out of 5 stars38 ratings	410.00
Daughter of the Deep	Rick Riordan	Soneela Nankani	11 hrs and 16 mins	05-10-21	English	4.5 out of 5 stars12 ratings	615.00
The Lightning Thief: Percy Jackson, Book 1	Rick Riordan	Jesse Bernstein	10 hrs	13-01-10	English	4.5 out of 5 stars181 ratings	820.00
The Hunger Games: Special Edition	Suzanne Collins	Tatiana Maslany	10 hrs and 35 mins	30-10-18	English	5 out of 5 stars72 ratings	656.00
Quest for the Diamond Sword	Winter Morgan	Luke Daniels	2 hrs and 23 mins	25-11-14	English	5 out of 5 stars11 ratings	233.00
The Dark Prophecy	Rick Riordan	Robbie Daymond	12 hrs and 32 mins	02-05-17	English	5 out of 5 stars50 ratings	820.00
Merlin Mission Collection	Mary Pope Osborne	Mary Pope Osborne	10 hrs and 56 mins	02-05-17	English	5 out of 5 stars5 ratings	1,256.00
The Tyrant's Tomb	Rick Riordan	Robbie Daymond	13 hrs and 22 mins	24-09-19	English	5 out of 5 stars58 ratings	820.00

In [8]:
```
# https://stackoverflow.com/questions/4289331/how-to-extract-numbers-from-a-str
import re
def extract_duration(x):
    if ('hrs' in x or 'hr' in x) and ('mins' in x or 'min' in x):
        hrs_mins = re.findall(r'\d+', x)
        duration = (int(hrs_mins[0])*60) + int(hrs_mins[1])
    elif ('hrs' in x or 'hr' in x) and not ('mins' in x or 'min' in x):
        hrs = re.findall(r'\d+', x)
        duration = (int(hrs[0])*60)
    elif ('mins' in x or 'min' in x) and not ('hrs' in x or 'hr' in x):
        mins = re.findall(r'\d+', x)
        duration = int(mins[0])
    else:
        duration = 0

    return duration

df["time"] = df["time"].apply(extract_duration)
df.rename(columns={"time": "duration_mins"}, inplace=True)
```

This function extracts the numbers from the strings in the time column, and converts them into runtime in minutes.

In [9]:
```
df[df["duration_mins"]==0]
```

Out[9]:

	author	narrator	duration_mins	release_date	language	stars	price
name							

In [10]:
```
df['release_date'] = pd.to_datetime(df["release_date"], format="%d-%m-%y")
```

In [11]:
```
df["language"].unique()
```

Out[11]:
```
array(['English', 'Hindi', 'spanish', 'german', 'french', 'catalan',
       'swedish', 'italian', 'danish', 'finnish', 'dutch', 'hebrew',
       'russian', 'polish', 'galician', 'afrikaans', 'icelandic',
       'romanian', 'japanese', 'tamil', 'portuguese', 'urdu', 'hungarian',
       'czech', 'bulgarian', 'mandarin_chinese', 'basque', 'korean',
       'arabic', 'greek', 'turkish', 'ukrainian', 'slovene', 'norwegian',
       'telugu', 'lithuanian'], dtype=object)
```

In [12]:
```
df["language"] = df["language"].apply(lambda x: x.capitalize())
```

Standardized language strings to all be capitalized.

In [13]:
```
df["stars"].replace({"Not rated yet": "0 out of 5 stars0 ratings"}, inplace=Tru
```

Standardized the "stars" column so the upcoming code can work on all rows.

In [14]:
```
df[["stars", "num_ratings"]] = df["stars"].str.split(" out of 5 stars", expand
```

I split the value of the old stars column to leave the star rating in one column, and create a new column with the number of ratings.

In [15]:
```
df["stars"] = df["stars"].astype("float")
```

In [16]:
```
df.head()
```

Out[16]:

name	author	narrator	duration_mins	release_date	language	stars	price	num_rating
Geronimo Stilton #11 & #12	Geronimo Stilton	Bill Lobely	140	2008-08-04	English	5.0	468.00	34 rating
The Burning Maze	Rick Riordan	Robbie Daymond	788	2018-05-01	English	4.5	820.00	41 rating
The Deep End	Jeff Kinney	Dan Russell	123	2020-11-06	English	4.5	410.00	38 rating
Daughter of the Deep	Rick Riordan	Soneela Nankani	676	2021-10-05	English	4.5	615.00	12 rating
The Lightning Thief: Percy Jackson, Book 1	Rick Riordan	Jesse Bernstein	600	2010-01-13	English	4.5	820.00	181 rating

In [17]:
```
df["num_ratings"].replace({' ratings': '', ' rating': '', ',': ''}, regex=True
```

In [18]:
```
df["num_ratings"] = df["num_ratings"].astype("int")
```

In [19]:
```
df["price"].replace({',': '', "Free": 0}, regex=True, inplace=True)
```

In [20]:
```
df["price"] = df["price"].astype("float")
```

In [21]:
```
df["price"] = df["price"].apply(lambda x: round(x/82.84, 2))
df.rename(columns={"price": "price_usd"}, inplace=True)
```

The price column was in Indian Rupees, so i converted it to USD using the current rate at Feb 21, 10:19PM UTC
Please note: these prices are likely not exactly accurate on Audible, but serve as a good basis for analysis.

In [22]:
```
df = df.iloc[:,[0,1,2,3,4,5,7,6]]
```

Reordered columns - having stars and number of ratings together makes the most sense.

In [23]:
```
df.head()
```

Out[23]:

name	author	narrator	duration_mins	release_date	language	stars	num_ratings	price_
Geronimo Stilton #11 & #12	Geronimo Stilton	Bill Lobely	140	2008-08-04	English	5.0	34	
The Burning Maze	Rick Riordan	Robbie Daymond	788	2018-05-01	English	4.5	41	
The Deep End	Jeff Kinney	Dan Russell	123	2020-11-06	English	4.5	38	
Daughter of the Deep	Rick Riordan	Soneela Nankani	676	2021-10-05	English	4.5	12	
The Lightning Thief: Percy Jackson, Book 1	Rick Riordan	Jesse Bernstein	600	2010-01-13	English	4.5	181	

In [24]:
```
df.duplicated().sum()
```

Out[24]: 2691

In [25]:
```
df.drop_duplicates(subset=["author", "narrator", "duration_mins", "release_dat
```

Some entries had a different number of ratings, so I dropped duplicates based on the other columns.

In [26]:
```
df.to_csv("audible_cleaned.csv")
```

Exported to .csv for future analysis!